

Final Model Selection

M5 Retail Demand Forecasting — Problem Statement 11

Phase 11 — the rule, the scorecard, the choice

NPN AIA Hackathon — St. Joseph's College of Engineering

Phase 11 — choosing on measured evidence against a rule fixed in advance. Generated 2026-08-14.

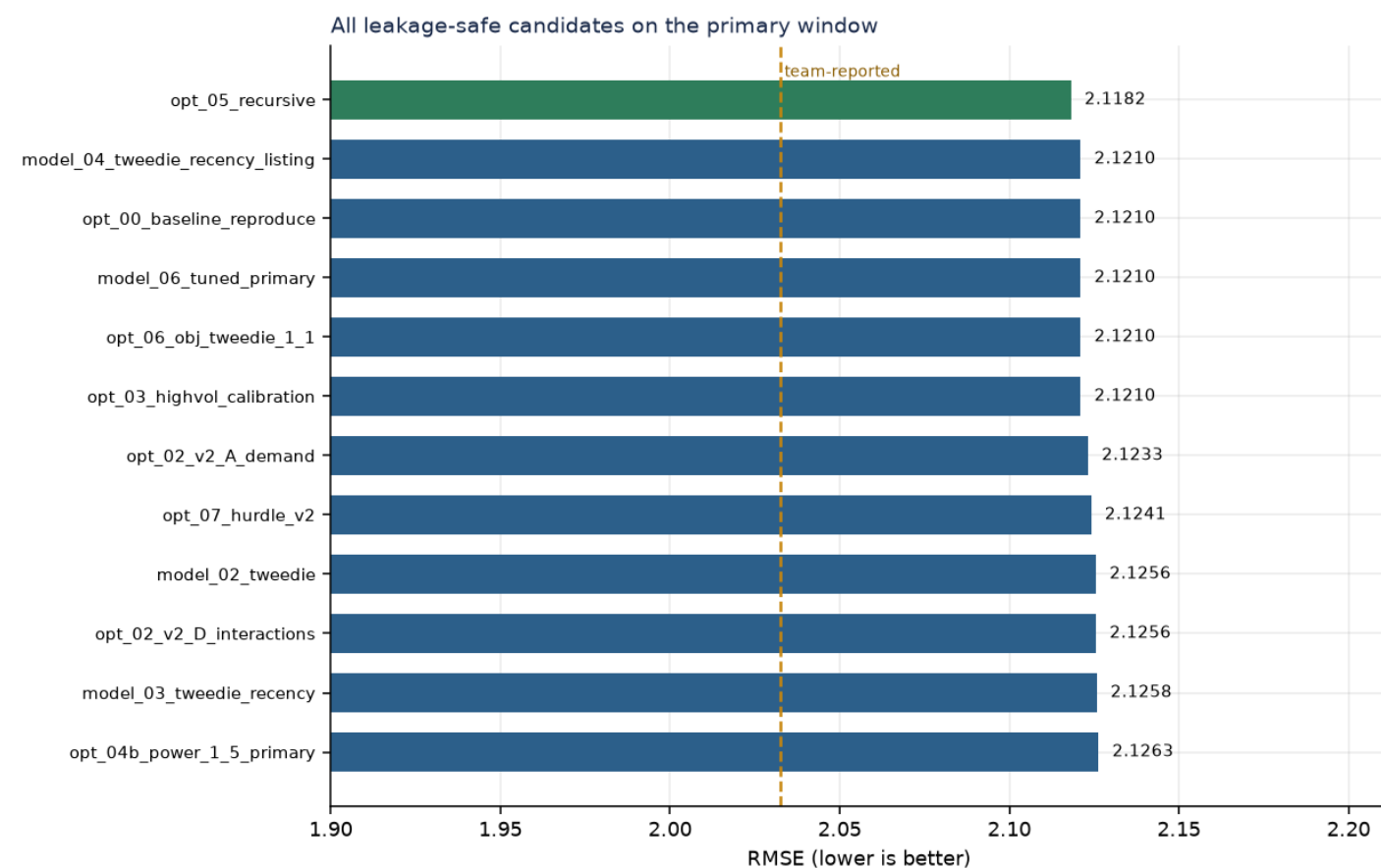
Terms. **RMSE** — average error with big misses punished far more heavily; lower is better. **MAE** — the plain average error. **Leakage** — letting the model see information that would not have existed when the forecast was really made. **Tweedie** — a loss function for data that is never negative and mostly zero. **Objective / loss function** — the definition of "wrong" that the model tries to minimise. **Inner window** — an earlier 28-day period we tune on, so the real scoring window stays untouched.

Every number here comes from an experiment that actually ran. Failed experiments are reported alongside successful ones — in this campaign most of them failed, and that is the finding.

The rule, set before the results were seen

- leakage-safe (a hard gate, not a ranking factor)
- lowest RMSE on the primary window
- MAE as tie-break — and as a veto if a trivial RMSE gain costs a lot of MAE
- robust across windows
- reasonable training time
- explainable
- novelty only if experimentally supported

Every candidate



Model	RMSE	MAE	Δ RMSE	Δ MAE
opt_05_recursive	2.1182	1.0717	-0.0029	+0.0398
model_04_tweedie_recency_listing	2.1210	1.0319	+0.0000	+0.0000
opt_00_baseline_reproduce	2.1210	1.0319	+0.0000	+0.0000
model_06_tuned_primary	2.1210	1.0319	+0.0000	+0.0000
opt_06_obj_tweedie_1_1	2.1210	1.0319	+0.0000	+0.0000
opt_03_highvol_calibration	2.1210	1.0319	+0.0000	-0.0000
opt_02_v2_A_demand	2.1233	1.0312	+0.0022	-0.0007
opt_07_hurdle_v2	2.1241	1.0300	+0.0031	-0.0020
model_02_tweedie	2.1256	1.0315	+0.0045	-0.0005
opt_02_v2_D_interactions	2.1256	1.0314	+0.0045	-0.0005
model_03_tweedie_recency	2.1258	1.0320	+0.0048	+0.0001
opt_04b_power_1_5_primary	2.1263	1.0289	+0.0053	-0.0030
model_09_tweedie_power_1_5	2.1263	1.0289	+0.0053	-0.0030
model_05_hurdle	2.1267	1.0324	+0.0057	+0.0004
opt_08_ensemble_tweedie_l1	2.1272	1.0128	+0.0062	-0.0191
opt_02_v2_C_price	2.1281	1.0307	+0.0071	-0.0012
opt_02_v2_all	2.1320	1.0313	+0.0110	-0.0007

Model	RMSE	MAE	Δ RMSE	Δ MAE
opt_02_v2_B_calendar	2.1327	1.0326	+0.0116	+0.0007
opt_06_obj_l2	2.1351	1.0388	+0.0141	+0.0069
opt_03_volume_weight_cap 5	2.1371	1.0335	+0.0161	+0.0016
opt_03_volume_weight_cap 3	2.1376	1.0336	+0.0165	+0.0016
opt_06_obj_poisson	2.1379	1.0350	+0.0169	+0.0031
model_01_lightgbm	2.1467	1.0411	+0.0256	+0.0092
opt_07_hurdle_v2_calibrate d	2.1822	1.0195	+0.0612	-0.0124
model_08_team_style_repr oduction	2.1835	1.0498	+0.0625	+0.0179
model_00_baseline_rolling_ mean_28	2.2430	1.0657	+0.1219	+0.0338
opt_06_obj_l1	2.2432	0.9591	+0.1221	-0.0728
model_00_baseline_rolling_ mean_7	2.2487	1.0683	+0.1276	+0.0364
model_00_baseline_season al_naive	2.6769	1.2440	+0.5559	+0.2120
model_00_baseline_last_va lue	2.8936	1.3730	+0.7725	+0.3411

Decision

Selected: opt_00_baseline_reproduce — RMSE 2.1210, MAE 1.0319.

The lowest-RMSE candidate (opt_05_recursive) improves RMSE by only 0.0029, which is inside the +/-0.013 window-to-window noise we measured, while costing +0.0398 MAE. That is not a real improvement, so the incumbent is retained.

Final forecast

- File: predictions\final_forecast_28day.csv — existing file verified, not regenerated (selected configuration unchanged)
- Mean forecast: 1.48287 units per series per day
- 30,490 rows, columns F1–F28, no NaN, no negatives, ids and order matching sample_submission.csv

No accuracy figure can be quoted for the forecast window itself (d_1942–d_1969) — no ground truth for it exists in any file. The validation result above is the only honest estimate.